

Machine-Learning Surrogate Optimization for Time-Based Consolidation in Last-Mile Parcel Delivery^{*}

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Abstract

Time-based consolidation (TBC) lets logistics service providers delay suitable parcels for batched delivery, trading customer waiting time against operational savings. While the principle is well known, its cost-service balance, its spatial signature, and its fleet-side benefits across heterogeneous carriers remain insufficiently quantified. We optimize TBC schedules on the Hanover Region case (seven LSPs, 1.26 million weekly parcels) using a machine-learning surrogate that couples Daganzo's continuum approximation with a LightGBM residual (2.95% mean absolute percentage error, MAPE, under postal-code-grouped cross-validation), embedded in a coordinate-descent search over a service-penalty P and a willingness-to-wait share θ . Cost savings reach 22.8% at the cost-optimal extreme, three carrier classes emerge with distinct sweet spots in $P \in [0.25, 0.75]$ € per parcel-day, and the Monday–Saturday fleet coefficient of variation drops by up to 54% in that range and by up to 78% at the most consolidated points. Rural postal-code areas with long stems, large service area, and few delivered parcels per stop benefit most. Out-of-sample VROOM re-routing confirms the surrogate is conservative, underestimating realized savings by 0.9–2.7 percentage points (pp) across four validated operating points.

Keywords: Temporal consolidation, Parcel delivery, Last-mile logistics, Vehicle routing problem, Delivery scheduling, Willingness to wait

Preprint note

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1. Introduction and Related Work

In the European Union, 78% of internet users purchased goods or services online in 2025 (Eurostat, 2025), illustrating how deeply parcel delivery is embedded in everyday consumption. Each online order creates a physical delivery task in the final segment of the parcel supply chain, commonly referred to as last-mile delivery (LMD), which is among the most expensive, inefficient, and environmentally burdensome parts of the supply chain (Olsson et al., 2019). For logistics service providers (LSPs), this intensifies existing pressures: LMD operations are already

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characterized by low productivity, restrictive delivery time windows, complex urban operating conditions, and high cost shares (Bachofner et al., 2022), reinforced by climate change, rapid urbanization, demographic change, and shifting demand patterns (Gruchmann et al., 2018).

At the same time, e-commerce has made delivery speed a visible part of the customer-facing service promise, so LSPs are expected to provide faster, more personalized, and more reliable services while prices and margins remain constrained (Vivaldini, 2022). However, not all customers prioritize the fastest possible delivery: Nguyen et al. (2019) distinguish price-oriented, convenience-oriented, and value-for-money-oriented segments, indicating that some customers may accept slower delivery if it lowers cost. This heterogeneity makes delivery timing a potential operational control variable. Temporarily storing eligible parcels and dispatching them jointly can reduce costs compared with direct delivery (Peng et al., 2023), and a some-day delivery option can reduce cost and vehicle requirements when customers accept a latest delivery day (Voigt et al., 2025); sustainability benefits, however, arise only if the temporal flexibility translates into better routing, lower fleet requirements, or reduced travel distances (Zeelenberg et al., 2026). Time-based consolidation (TBC) packages this trade-off into a delivery concept in which eligible customers accept a longer delivery window, for example in exchange for a lower price (Naeimian et al., 2026), so that eligible shipments are temporarily held and grouped into batched deliveries within a predefined service promise. TBC therefore moves away from a uniform immediacy logic toward a differentiated service design in which customers retain choice while LSPs gain temporal flexibility.

This paper investigates TBC as an operational response to the tension between customer expectations, LSP constraints, and sustainability objectives. From a planning perspective, the core task is to assign eligible parcels to delivery days at the postal-code-area level such that delivery density increases, additional waiting time remains limited, and weekday demand peaks or uneven fleet use are avoided. Because each schedule induces a different demand pattern and thus different delivery routes, direct evaluation would require solving many Vehicle Routing Problems, which is computationally infeasible given the VRP’s NP-hardness (Lenstra and Kan, 1981). We therefore propose an ML-based surrogate-optimization framework for identifying cost-efficient TBC schedules in a large-scale case study of the Hanover Region in Germany. The framework combines a postal-code-area scheduling model with a routing-cost surrogate trained on VRP solutions; building on ML-supported combinatorial optimization (Bengio et al., 2021) and data-driven extensions of analytical LMD route-distance approximations (Figliozzi, 2008; Kirubakaran et al., 2026), it enables efficient schedule search across adoption scenarios. We make three contributions: a hybrid Daganzo–LightGBM routing-cost surrogate that couples an interpretable continuum backbone with a learned residual; a scalable optimization of spatially and provider-differentiated TBC delivery-day patterns; and empirical evidence on where TBC pays off and how the adoption level governs its fleet effects. Routing and demand simulation enter only as label generation and out-of-sample evaluation, so the framework is a surrogate-assisted delivery-schedule optimization rather than a simulation study.

2. Methodology

Our framework used weekly parcel-demand patterns generated with our own HAGRID demand model (Bienzeisler et al., 2026), covering seven LSPs in the Hanover Region (about 1.26 million parcels and 1.91 M€ of baseline operating cost per week). As Figure 1 shows, demand varies strongly between dense urban and sparse rural postal-code areas, peaks midweek, and is lowest on Saturday. To make repeated routing evaluations tractable, we decomposed the study area into postal-code-level routing subproblems while preserving the demand, spatial, and

depot-access characteristics relevant for delivery cost. The pipeline structure is summarized in Figure 2.¹

The unit of analysis is a *cell*, defined as an LSP–postal-code-area pair assigned to the provider’s serving hub and described by daily demand, area size, stop structure, and depot-access distance. The decision variable is the weekly *delivery pattern*, i.e., the service weekdays for that cell; parcels from non-service days are held until the next scheduled service day. LSP operating costs are route-dependent. Following open-data-based LMD modeling (Hörl et al., 2025), we generated routing-cost labels with VROOM (Coupey et al., 2024) based on Valhalla-derived travel times and distances on OpenStreetMap road geometry.² Recurrent congestion was approximated by applying an empirically derived correction factor to free-flow travel times. The current daily-delivery operation served as the baseline for all consolidation scenarios.

2.1. Cost approximation and machine-learning surrogate

Solving a full VRP for every candidate schedule is computationally too costly, so we approximated operational cost with a continuous model based on Daganzo’s distance formulation (Daganzo, 1984) and corrected it with a data-driven term. For area z on day d , route distance and cost are

$$\hat{D}_{z,d} = 2m_{z,d}r_z + \beta\sqrt{n_{z,d}A_z}, \quad \hat{C}_{z,d} = c_d\hat{D}_{z,d} + c_t(\hat{D}_{z,d}/v + sn_{z,d}) + c_fm_{z,d}, \quad (1)$$

with $m_{z,d} = \lceil p_{z,d}/Q \rceil$ tours (parcel volume $p_{z,d}$, vehicle capacity Q), average depot distance r_z , $n_{z,d}$ stops over area A_z , network/shape factor β , and distance/time/fixed-cost rates c_d, c_t, c_f , speed v , and per-stop service time s . Consolidation lowers cost by raising delivery density while the capacity term penalizes over-consolidation. Such approximations reproduce the first-order scaling but are point-inaccurate, as they abstract from the road network, depot position, and demand clustering, normally requiring manual recalibration (Figliozzi, 2008). Instead, we used Equation (1) as a physically interpretable *backbone* and learned its deviation from solver cost,

$$\tilde{C}_{z,d} = \alpha\hat{C}_{z,d} + g(\mathbf{x}_{z,d}), \quad (2)$$

where α is a global calibration factor and the structured residual $g(\cdot)$ is a LightGBM ensemble (Ke et al., 2017), chosen because gradient-boosted trees handle tabular nonlinearities and feature interactions well, and expose feature importance for interpretation (Grinsztajn et al., 2022). The

¹The implementation of the proposed framework is available at: <https://github.com/TUBS-IVS/vroom-valhalla-lmd-hannover>

²<https://github.com/valhalla/valhalla>

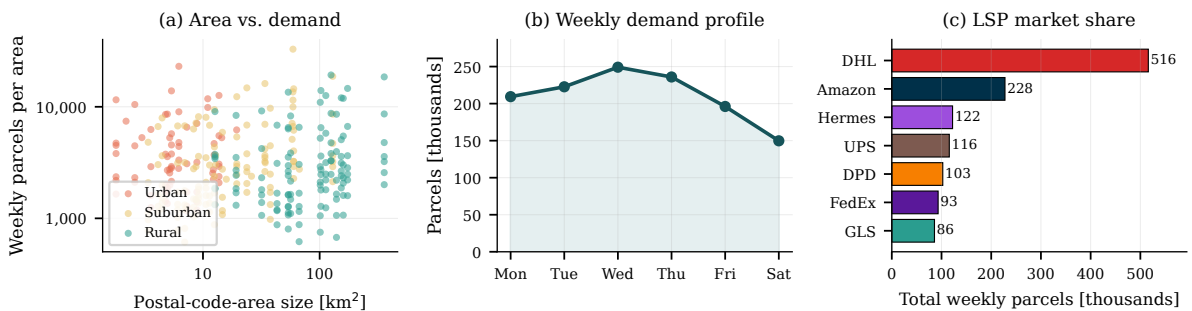


Figure 1: Input data from HAGRID (a) Postal-code-area size vs. weekly volume by region type; (b) weekly demand profile; (c) LSP market share.

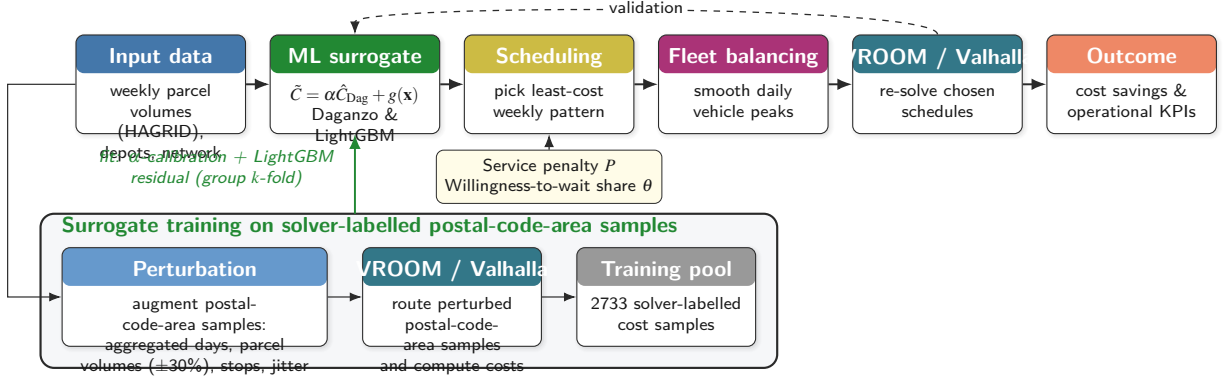


Figure 2: Surrogate-based optimization and validation pipeline for TBC.

scalar α was calibrated once as the median ratio of solver cost to the raw Daganzo prediction across the training pool, and $g(\cdot)$ then learned the structured remainder.

The feature vector $\mathbf{x}_{z,d}$ comprised 25 demand and geometry descriptors derived ex ante from the data, following the taxonomy of Akkerman and Mes (2025) (parcel volume, stops, service-area size, depot distance, parcels per stop, the load factor $p_{z,d}/Q$, and per-stop demand statistics), with VROOM/Valhalla route costs as training targets. To cover the consolidation cost range rather than only baseline operations, we augmented the pool with synthetic cell-day samples obtained by demand scaling, stop dropout, coordinate jitter, and multi-day aggregation, the last reproducing the elevated per-day volumes of batched delivery. Generalization to both sparse rural and dense urban areas was assessed by group k -fold cross-validation with folds grouped by postal-code area.

2.2. Schedule optimization and fleet balancing

For each *cell* (each LSP, postal-code area pair), we enumerated all weekly delivery patterns feasible under a maximum holding time of $H_{\max} = 3$ days over the six-day operating week, yielding 39 distinct admissible patterns; the same candidate set is evaluated for every cell, while the resulting demand, waiting time, and cost differ by cell. For a pattern σ , a delivery day served its own demand plus the parcels held from the preceding non-delivery days, while non-delivery days carried no consolidated tour. The average customer waiting time $\bar{w}(\sigma)$ is the mean delay over the six weekdays between parcel arrival and the next scheduled service day under a cyclic interpretation of σ . Cells served by the same depot share a pooled daily delivery tour for the non-consolidated share of their demand, so the surrogate cost of a cell is not separable from the patterns chosen by its hub neighbors. We therefore solved the selection by coordinate descent, run independently within each provider's hub network: at every update the focal cell chose the pattern that minimized the weekly surrogate cost evaluated against the current schedules of all other cells at the same hub, augmented by a linear service penalty,

$$\sigma_z^* = \arg \min_{\sigma \in \mathcal{S}} \left[\sum_d \tilde{C}_{z,d}(\sigma \mid \sigma_{-z}) + P \theta_z p_z \bar{w}(\sigma) \right], \quad (3)$$

where $\mathcal{S}$ is the set of feasible patterns, σ_{-z} the fixed patterns of all other cells at the focal cell's hub, p_z the weekly parcel volume, and θ_z the share of parcels eligible for consolidation in the cell. Following a business-prioritized power law of the system-wide adoption θ , business recipients opt in earlier than private ones, so each cell's θ_z reflects its own composition while the volume-weighted regional average equals θ ; only this willing fraction is consolidated, the rest delivered conventionally. Coordinate descent iterates with multiple random restarts until

no single-cell update improves the hub objective. Cells sharing a hub are coupled only through the pooled residual tour, and each single-cell update is evaluated against the full hub cost; the procedure is therefore coordinate descent on a common hub objective, not a game between self-interested cells, and its fixed point is a coordinate-wise local optimum—equivalently, a Nash equilibrium of an identical-interest game. Across five random restarts the objective agreed to within 10^{-12} , so the optimum is stable at the reported operating points, although global optimality is not guaranteed. The optimization runs independently within each LSP’s own hub network; no parcels, routes, or vehicles are shared across providers, and system-wide figures are obtained solely by aggregating the independently optimized provider solutions. The service penalty P is a shadow price (Boyd and Vandenberghe, 2004) of waiting, i.e. the marginal cost the operator trades for one additional parcel-day of delay, and enters (3) as a steering term only, not as a cost component, which keeps the cost-service trade-off continuous across the grid. We swept (P, θ) over eight P levels from 0 to 10 €/p/d and eleven θ levels from 0 to 1 in steps of 0.1.

A cost-optimal selection alone can concentrate deliveries on a few weekdays, producing peaks that are hard to staff under realistic subcontractor and roster constraints. A two-stage post-processing therefore preserves each cell’s chosen delivery frequency: a per-depot greedy local search reassigns weekdays to flatten each depot’s peak, accepting only swaps that keep total weekly cost within 5% of the cost-optimal value, and a subsequent smoothing stage, applied separately within each provider, redistributes weekday assignments across that provider’s hubs to flatten its Monday-to-Saturday fleet.

3. Results and Discussion

3.1. Surrogate model: quality and validation

Under group k -fold cross-validation with folds grouped by postal-code area, so that no area appeared in both training and validation, the Daganzo-LightGBM hybrid (Daganzo-LGB-Hybrid) achieved 2.95% MAPE ($R^2 = 0.997$). We kept the Daganzo cost parameters at standard operational values and calibrated the global cost level through the factor α , defined as the median ratio of solver cost to Daganzo cost ($\alpha = 1.34$). A single global α suffices: replacing it with per-LSP factors did not improve out-of-fold accuracy (3.2% vs. 2.9% MAPE), because the LightGBM residual already absorbs provider-specific scale. Figure 3 reports the out-of-fold accuracy in three stages: the uncalibrated approximation underestimated cost by about 26% (panel a); the global α calibration removed this systematic bias and reached 9.7% MAPE (panel b); and the learned LightGBM residual captured remaining effects related to capacity thresholds, stop density, and depot distance, reducing the error to 2.95% with near-zero bias (panel c). Overall, among the benchmark models tested in this study, including both statistical and ML approaches, the Daganzo-LGB-Hybrid achieved the best predictive performance. The banded structure in Figure 3 reflects the integer tour count $\lceil p/Q \rceil$, which quantizes the Daganzo backbone into cost bands, together with families of augmented samples sharing a postal-code area; it is a property of the training pool rather than a fit artifact, and the postal-code-grouped folds keep all samples of an area on the same side of every split.

3.2. Schedule optimization and the cost–service trade-off

Figure 4 traces the optimized delivery-day distribution across the 312 postal-code areas as θ rises; we summarize each scenario by the mean delivery frequency $\bar{f}$, the unweighted average number of weekly delivery days across cells. At $\theta = 0$ every area is served daily at any penalty, recovering the baseline. As θ rises, P sets the pace: at $(P, \theta) = (0, 1)$, 97.4% of areas adopt the

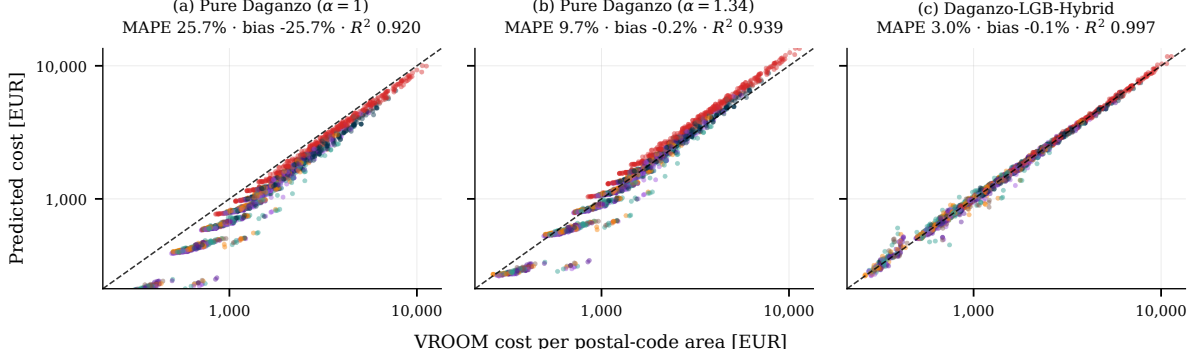


Figure 3: Predicted vs. VROOM routing cost under group k -fold validation by postal-code area. Points are coloured by LSP. Panels show (a) the raw Daganzo approximation, (b) the globally α -calibrated Daganzo approximation, and (c) the final Daganzo-LGB-Hybrid surrogate.

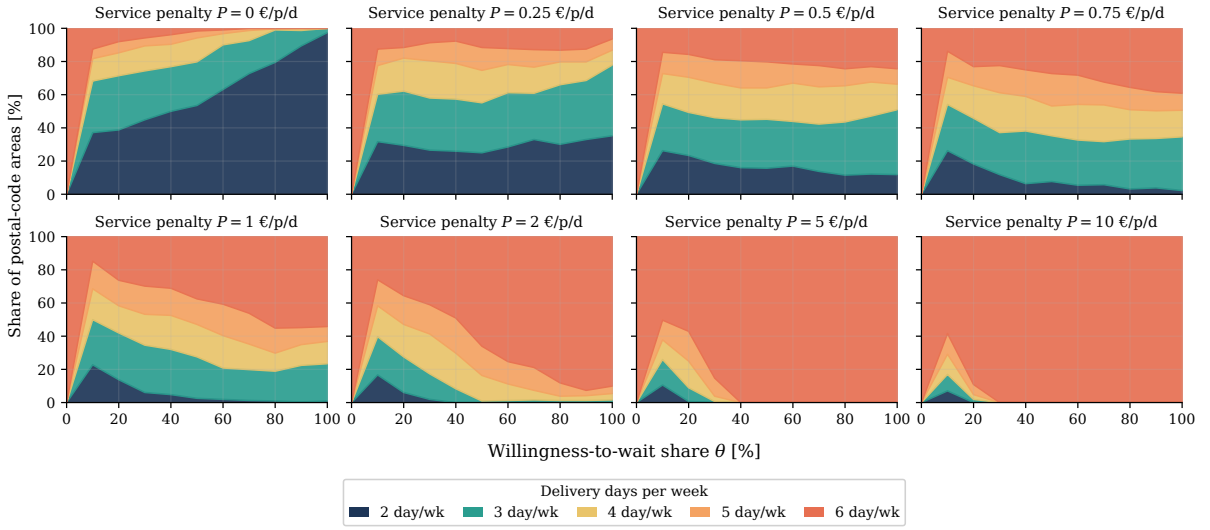


Figure 4: Delivery-frequency mix across postal-code areas as a function of the willingness-to-wait share θ at different service penalties P .

minimum two-day pattern ($\bar{f} = 2.03$); the remaining 2.6% are high-volume urban cells whose two-day demand would breach vehicle capacity, so the Daganzo capacity term forces an extra day. The mode shifts upward through $P = 0.5$ ($\bar{f} = 3.95$, 24% daily at the capacity floor) and $P = 1$ ($\bar{f} = 4.93$, 54% daily); for $P \geq 5$ and $\theta \geq 0.3$ the system reverts to fully daily delivery. This reversion is non-monotone in θ : at $\theta = 0.1$ even $P = 10$ still leaves 42% of areas non-daily ($\bar{f} = 5.05$), with the daily share rising only from $\sim 14\%$ at $P \leq 1$ to 58% at $P = 10$. Two mechanisms compound. First, the penalty scales as $P \cdot \theta_z \cdot p_z \cdot \bar{w}$, so the effective pressure $P \cdot \theta$ at $(10, 0.1)$ matches that at $(1, 1)$ (1 € per parcel-day). Second, hub-bundling lets rural and low-density suburban cells share a depot tour with neighbors, so the routing saving on the 90% standard-delivery parcels covers the small penalty on the willing 10%. The balance is fragile because θ enters linearly: at $\theta = 0.2$ only 11% remain non-daily at $P = 10$, and by $\theta = 0.3$ ($P \cdot \theta = 3$) the offset is exhausted.

Figure 5 reports cost and fleet outcomes across the (P, θ) grid. Cost saving relative to the daily-delivery baseline (a, b) peaks at 22.8% in the fully consolidated regime $(P, \theta) = (0, 1)$ and decays monotonically with P ; fleet balancing costs only 0.3–0.6 pp of saving in the high-saving region, so schedule selection captures the bulk of the benefit while balancing trades a marginal

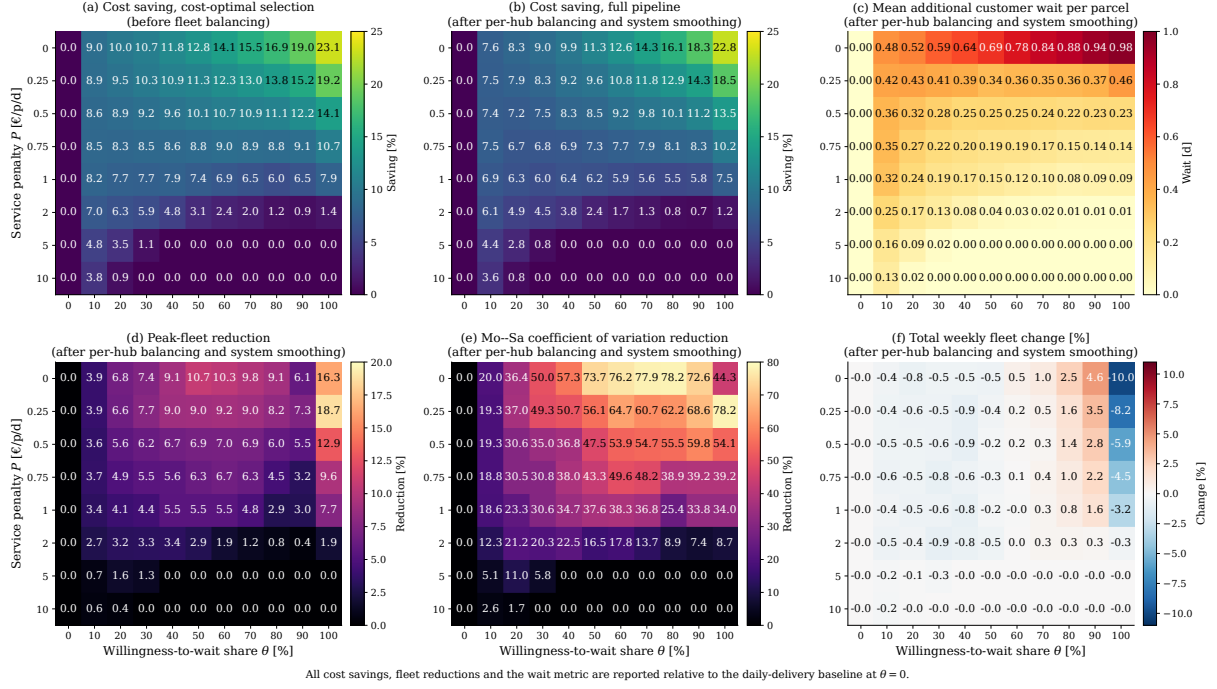


Figure 5: Total cost saving (a, b), parcels-weighted additional customer wait (c), and fleet metrics (d-f) over the service-penalty P and willingness-to-wait θ grid, reported relative to the daily-delivery baseline.

cost increment for fleet smoothness. The customer-side cost grows much more slowly than the saving decays: the parcel-weighted additional wait drops from 0.98 d at $P = 0$ to 0.46 d at $P = 0.25$ and to 0.23 d at $P = 0.5$ (c), more than halving with just 4.2 pp of saving lost over $[0, 0.25]$ before the marginal cost of further wait reduction triples from 7.9 to 23.2 pp/d over $[0.25, 0.5]$. On the fleet side, the Monday–Saturday coefficient of variation falls by 54% at $(0.5, 1)$ and by up to 78% across the grid (e), and peak fleet by 12.9% at $(0.5, 1)$ with 8–14% across the cost-friendly region (d), but the total weekly fleet (f) only drops meaningfully at full adoption (-10% at $P = 0$, -8.3% at $P = 0.25$, -5.9% at $P = 0.5$) and can grow by up to 4.6% at intermediate θ . The reason is structural: as long as some parcels remain in conventional delivery, the operator must run a daily conventional tour in parallel with the batched delivery days, so the peak and CV gains of (d) and (e) reflect a redistribution of vehicle requirement across weekdays rather than an absolute reduction.

The per-LSP operating point P^* is identified as the chord-distance knee of each carrier’s (saving, wait) Pareto front at $\theta = 1$ (Figure 6, a), i.e. the point with the largest orthogonal distance to the chord between the cost-minimum and wait-minimum extremes, beyond which the saving lost per further day of wait reduction grows sharply. This groups the seven operators into service-bound (Amazon, DHL) at $P^* = 0.25$, hybrid (FedEx, Hermes, UPS) at $P^* = 0.5$, and cost-aggressive (DPD, GLS) at $P^* = 0.75$. Which areas can consolidate is governed by the local routing geometry: the per-area saving correlates strongly with parcels per drop-site ($\rho = -0.72$), hub distance ($\rho = +0.53$), and area size ($\rho = +0.31$); dense urban cells with short stems and high parcels-per-stop gain little from waiting, while sparse rural cells with long stems benefit most. The same mechanism caps the service-bound class at a 10% per-postal-code median saving while the hybrid and cost-aggressive classes reach 22–24% at their P^* (b), and at low adoption ($\theta \leq 0.3$) it even pushes 35–44% of service-bound cells into negative saving, since the parallel conventional tour for the 70–90% non-willing parcels exceeds the batching gain for cells near the capacity floor. At each P^* and full adoption, rural areas save a median of

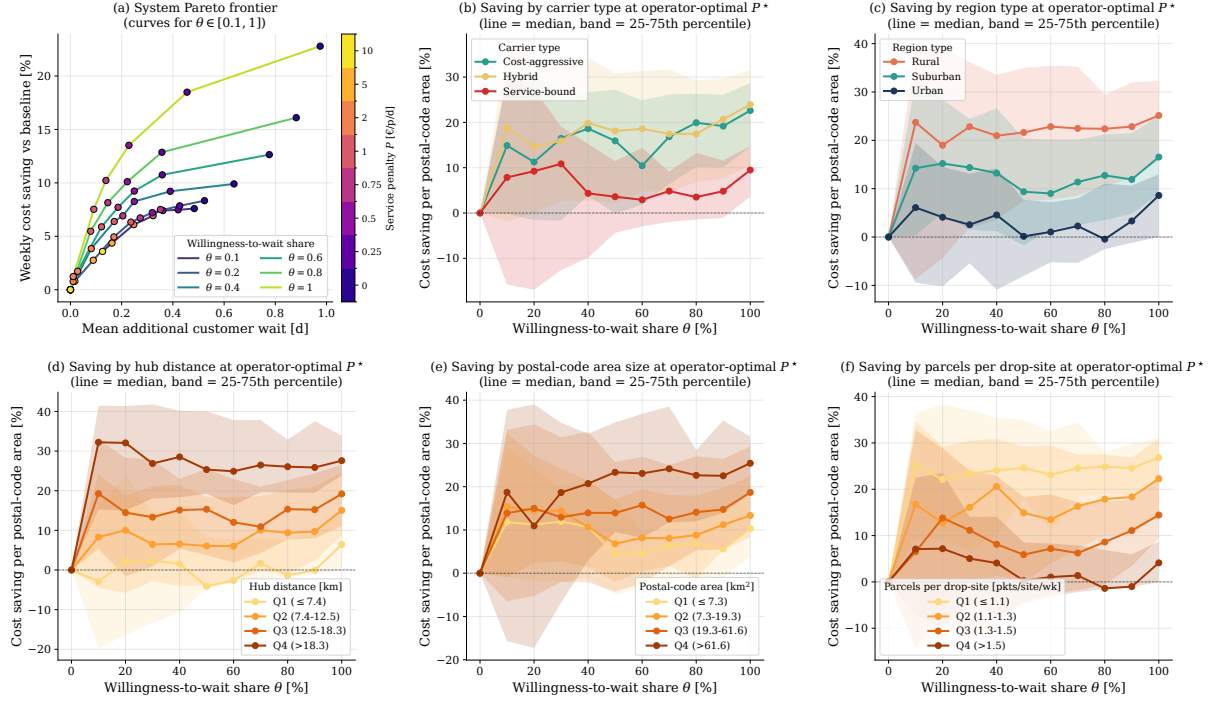


Figure 6: System Pareto frontier (a) and median per-postal-code cost saving by carrier type (b), region type (c), hub distance (d), postal-code area (e), and parcels per drop-site (f) over the willingness-to-wait share θ at each LSP’s optimal service penalty P^* .

25% versus 9% for urban areas (c), and the structural splits (d–f) confirm this ordering: median saving rises from 8% to 28% with hub distance and from 10% to 25% with area size, while it falls from 27% to 4% with parcels per drop-site between quartiles. TBC therefore pays most where delivery is sparse and far from the depot. This efficiency pattern, however, creates an equity concern: service quality between urban and rural areas should be maintained (Pereira et al., 2017), especially since rural and suburban deliveries account for a substantial share of LMD activity (Bienzeisler et al., 2026).

As a final step, we re-routed the optimized schedules with VROOM as an out-of-sample validation of the surrogate. At $\theta = 1$ and four validated operating points $P \in \{0, 0.25, 0.5, 0.75\}$, the Daganzo-LGB-Hybrid matches the solver-actual system-wide cost saving within 0.9–2.7 pp: 22.8% predicted vs. 23.7% actual at $P = 0$, 18.5% vs. 19.8% at $P = 0.25$, 13.5% vs. 15.6% at $P = 0.5$, and 10.2% vs. 13.0% at $P = 0.75$. The surrogate is consistently conservative, slightly understating the achievable savings across the scenario space, with the gap widening at higher penalties.

4. Conclusion

We presented an ML-based surrogate optimization framework for time-based consolidation in last-mile parcel delivery on a case study of seven LSPs, 312 postal-code areas, and 1.26 million parcels per week in the Hanover Region. The Daganzo-LightGBM hybrid surrogate achieved 2.95% MAPE under postal-code-grouped cross-validation, enabling a full (P, θ) sweep. The operationally efficient range lies between $P = 0.25$ and $P = 0.5$ €/p/d, capturing 13.5–18.5% cost saving at $\theta = 1$ (up to 22.8% at the cost-optimal extreme $P = 0$) and a 54% reduction in the Monday–Saturday fleet coefficient of variation, while the absolute weekly vehicle count only falls at full adoption. Per-LSP chord-distance optima sort the operators into

service-bound, hybrid, and cost-aggressive classes, with savings driven by long stems, large postal-code areas, and few parcels per drop-site. For practice, this suggests offering TBC first in rural, depot-distant areas with few parcels per stop and differentiating the service spatially and by provider rather than uniformly; because a low adoption share forces conventional and consolidated tours to run in parallel, an absolute fleet reduction materializes mainly at high adoption, whereas the fleet-smoothing benefit is already captured at moderate penalties for only 0.3–0.6 pp of forgone saving. TBC therefore emerges as a particularly attractive lever for rural regions, where the structural gains are largest. Validation relies on VROOM as a routing benchmark rather than ground truth, and our per-postal-code-area decomposition rules out inter-area consolidation that could further amplify TBC gains. Future work will lift this constraint by jointly routing neighbouring postal-code areas within the solver’s tractability limits.

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